SEDAR

Southeast Data, Assessment, and Review

SEDAR 98

Gulf Red Snapper

SECTION IV: Research Recommendations

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1. DATA WORKSHOP RESEARCH RECOMMENDATIONS

1.1 LIFE HISTORY RESEARCH RECOMMENDATIONS

The LHWG drafted the following recommendations:

- During the SEDAR98 data workshop, the LHWG examined how changes in the spatial distribution of port samplers over time affected length or age sample representativeness, considering regional differences in size-at-age observed among the three GOA regions. We recommend that sample collection methods be examined to determine if different dockside sampling protocols (e.g., quota sampling, length-bin sampling, or proportional sampling) may have biased length or age distributions of sampled fish compared to the landings.
- Updating database infrastructure would allow for further investigation into sample representativeness for samples collected during the mid and early data periods (i.e., prior to 2011), specifically comparisons of length distributions for all samples received versus samples processed for ageing.
- The current subsampling protocol for red snapper is based on 5-year average landings by grid and is excessively laborious and time consuming. Evaluate the current otolith subsampling protocol and provide alternatives to streamline the process.
- As recommended in SEDAR74, additional personnel and database infrastructure resources are still needed to manage large life-history datasets that now exhaust the capabilities of standard computers.
- Expanding routine biological sampling in the East region, especially in the southernmost part of the west coast of FL, likely would reduce uncertainty in parameters estimates for

the East region, where sample sizes are relatively low compared to the West or Central region.

- Evaluate the current otolith subsampling protocol used by the NMFS PC Lab to improve efficiency in sample processing (i.e., sample selection for processing as well as minimum sample sizes needed to maintain age data quality/integrity)
- Evaluate the sampling design for observer programs to increase the number of age samples collected, especially for the COM LL fleets.
- Continue to evaluate next-gen technologies (e.g., FT-NIRS, epigenetics) to improve age sample processing efficiency for high-volume sample fleets (e.g., COM HL) while maintaining age data quality/integrity.
- As recommended in SEDAR74, increase sampling of sublegal fish through fishery independent surveys and the shrimp observer program to better estimate maturity and fecundity of smaller individuals, as well as samples through tournament intercepts to better estimate batch fecundity of larger/older females, especially in the East region where sample sizes are especially low.
- We recommend that growth functions be fit externally to age-at-length data separately for all three regions during future SEDAR assessments. Growth parameters are not as well estimated for fish from the East region compared to the other two regions, and additional age-at-length data may increase the precision of growth parameter estimates in future assessments such that all three regions may require separate growth parameters.

1.2 COMMERCIAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

Although the Work Group focused on understanding and evaluating Red Snapper bycatch methods and estimates, the working paper SEDAR98-DW-25 outlines current efforts by SEFSC to improve bycatch estimation in the future. These include:

- outfitting the commercial shrimp fleet with modern, tamper-proof GPS tracklog devices;
- applying electronic technologies to enable observers to collect size information on all fish species captured in fish trawls;
- modifying the allocation of observer sampling effort to better incorporate net configuration and diurnal period;
- scoping of field experiments for calibrating SEAMAP and commercial shrimp trawls.

1.3 RECREATIONAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

1. Investigate potential options to smooth the ratios used to delineate historic recreational landings by region and fishing mode across time blocks.
2. Develop a Gulf wide interstate calibration between all states
    a. Evaluate future Creel surveys in MS and AL against LA Creel survey to determine if surveys are in equivalent units (Are the landings streams 1 to 1?)
    b. If creel surveys in LA, MS, and AL are in equivalent units, evaluate whether a Creel survey unit is equivalent to a SRFS unit
    c. Compare TPWD to the other Gulf state units
3. Evaluate the for-hire partitioning in the MRIP 1981-1985 years
4. Continue investigations into iSnapper/Texas Hunt and Fish app as a suitable source of discard and landings estimates for TX.
5. Evaluate additional sources of mortality from shark and marine mammal predation.
6. Research and compile Gulf hurricanes areas and dates over time.

1.4 MEASURES OF POPULTAION ABUNDANCE RESEARCH RECOMMENDATIONS

- Consider incorporating artificial reef video with natural bottom video data

2. ASSESSMENT PROCESS RESEARCH RECOMMENDATIONS

Recommendations for considerations of future research are provided below in no particular order of priority.

Recreational Landings and Discards data

Investigate influence of depredation as a contributor to discard mortality and its significance on observed discard data used in the assessment.

Additional Data Needs

Investigate the impact of using state survey derived landing statistics on the assessment model.

Use simulation studies to investigate the effectiveness of composition data sources (i.e., length, age and conditional age-at-length) and quality on the estimation of growth parameters in a red snapper-like species assessment.

Environmental Influences

Investigate non-anthropogenic changes in the Red snapper system that may be contributing to recruitment fluctuations in time and space. Continue to gain insight on recruitment apportionment for model predictions.

Modeling Platform Updates

Investigate risk associated with switching modeling platforms as newer more modular platforms are developed (i.e., FIMS).

3. REVIEW PANEL RESEARCH RECOMMENDATIONS

The following are suggested research directions for achieving the goal of overall improvement to the assessment and for improvement in the data and modeling. Note that these are recommendations for improving future assessments, rather than modifying the current one. With ***adequate human analytical resources,*** most could be achieved within a six month criterion. Longer term studies are noted.

- Spatial models should be used for index standardization. All abundance indices are currently calculated using a design-based approach; however, the spatial coverage of the different surveys has changed over time, and in such cases, spatio-temporal models may produce different results, as they can account for spatial correlation structure in addition to covariate effects to "fill in" gaps in spatial coverage. Furthermore, while depth effects have been included in the current model, they have only been treated as linear, whereas in practice the depth effect is often dome-shaped. It is therefore recommended that spatio-temporal models (e.g., using sdmTMB, mgcv/GAM, etc.) be explored, and that depth be incorporated as a covariate using a quadratic or smooth (non-linear) term rather than a linear one.
- The approach used to determine the MSY proxy should be revisited. Previous analyses established this value; however, if the species' life history parameters have since changed, a corresponding change from an SPR-based proxy of 26% would be expected. A review is therefore warranted, along with a broader reconsideration of reference points generally, particularly if parameters such as steepness are also revised.
- The uncertainty (CV) associated with landings data should be accounted for as accurately as possible within the assessment model. The current approach, while workable, should not be regarded as final and should be revisited and refined as improved methods or updated data become available.
- Forecasting using a weighted-average approach could be valuable for parameters such as recruitment, where the most recent estimate can be highly uncertain. A weighting procedure that accounts for uncertainty (e.g., weighting by 1/CV or 1/SD^2) would help address this issue.
- The choice to remove recreational time-varying discard mortality may warrant reconsideration when data availability permits. Anecdotal reports from stakeholders suggest fishing is occurring in progressively deeper waters in the Central region. Given the considerable magnitude of recreational discarding within this fishery, future

assessments would benefit from data informing the context under which catch and release occurs, as well as whether this context is changing over time.

- Consideration should be given to combining retention or selectivity parameters to simplify the model. While the original parameterization was well specified based on management measures and historical changes in the fishery, given current data limitations, this specification creates confounding in parameter estimates and creates instability in model estimates (as shown in the jitter analysis).
- Internal estimation of growth parameters remains an open research question. As noted elsewhere in this report, estimating growth externally often disregards the effect of selectivity, whereas internal estimation accounts for it; this distinction matters because whether or not selectivity is accounted for directly affects the estimate of *k*, and consequently the perceived trend and status of the stock. In a sensitivity run freeing only *k* (with other growth parameters fixed), the total negative log-likelihood improved by approximately 700 units relative to the base model, though four parameters—differing from those in the base run—remained at their bounds. Ideally, both L∞ and *k* would be estimated internally, but selectivity confounds their simultaneous estimation in this assessment. Further research is needed to determine best practice here—specifically, whether to fix one growth parameter while freely estimating the other, or to rely on external estimates for both (longer term).
- Relatedly, the age-length (AL) data should be corrected, as the dataset currently includes anomalous records (e.g., old fish with unexpectedly small body sizes). Prior to the next assessment, these records should be checked for potential input data errors (e.g., typographical errors). This is important as it may fix the inability of the model to estimate growth internally in the central area (which is also the place where we see fish with unexpected length at age attributes, especially from the charter boat). This should be a priority as it is "low" cost but with potentially high reward.
- Further research is needed on best practices for selecting Sigma R and its relationship with steepness, M and other factors. This is not solely an issue with red snapper. It is an issue confronting many stock assessments. This could be addressed through simulation studies (longer term).

- For any future recreational composition data derived from port sampling, the input sample size should be based on either the number of port sampling days or the number of distinct boats sampled, or alternatively the method proposed by Jim Thorson in the Stock Synthesis User Manual, Version 3.30.25.1. The rationale for using port sampling days is as follows: if boaters are assumed to concentrate in a similar area on a given day, each sampling occasion will reflect a somewhat different spatial distribution, potentially due to changes in fish distribution; in this case, the number of port sampling days would serve as a reasonable proxy for input sample size. Alternatively, if each boater is assumed to have a preferred fishing location, tracking the total number of distinct boaters

may be more appropriate—though in this case, it should be noted that many boats may fish in the same location, which would affect the independence of samples.